L5 - Vibration Testing of Beams

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All information documented and discussed in the following report is under the sole authorship of Anuranan Bharadwaj, Kalkamanali Satvaldy, Ryan Orr, James Nguyen. No part of this report has been published or submitted for publication.

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Abstract

When choosing materials used for manufacturing, a material's mechanical properties, such as its ability to withstand different loads, are one of the key characteristics used to determine material viability. However, under specific conditions, materials can fail at much lower loads; one such example is when the load is applied periodically, matching the material's natural/resonance frequency. An example of a catastrophic outcome is the collapse of the Tacoma Narrows Bridge in 1940, where wind applied periodic loads closely matching the bridge's resonance frequency, causing aeroelastic flutter. This caused the bridge to sway and undulate until its structural elements began to fail and eventually collapsed into the water below. This same phenomenon can be seen in aerospace with wing flutter or during air turbulence. One of the methods to prevent this is through vibration testing, experimentally determining an object's resonance frequency, identifying what conditions that could cause such frequencies, and altering the object's design to avoid such conditions while in use. Vibration testing is the process of applying known periodic loads to an object using a machine that vibrates at a set frequency, increasing the frequency until the object begins to exhibit large displacements, this frequency being that object's resonance frequency. The experiment used an electrodynamic shaker, which turns electrical current from the sine wave generator into adjustable mechanical forces. In addition, a strobe light was also used to help visualize the sine wave by seemingly bringing the object to rest while in motion, showing where the nodes are, and an oscilloscope was also used to find the period of the sine wave. Nodes are points on an oscillating object where there is no movement or where the oscillating object is in line with its original position. The objects tested were two Al-6061-T651 beams, one measuring 761 mm and the other measuring 1275 mm, with similar cross-sectional areas of 5.05 mm and 5.01 mm in height, 25.55 mm and 25.52 mm for the bases, respectively. Noting down the measured frequencies from the strobe light, and the distances of the nodes from the end of the beam attached to the electrodynamic shaker. Using these values, a theoretical value for resonance frequency can be calculated, and when compared to the experimental results measured from the sine wave generator, strobe light, and the oscilloscope, the maximum percent difference was over 11% for all three, only for the shorter aluminum beam, under 8% for the longer beam, both values decreasing as the mode decreases. Repeating this but comparing just the experimental values to one another comes out with a maximum of 13% which stands out as there is no trend that would explain this, pointing to the possibility of lower modes being unreliable. Finally, comparing the measured node distances to the expected node distances shows a maximum of 3% difference for both beams, mode 3 for the short beam, and mode 4 for the long beam.

Introduction

Physical systems must undergo both static and dynamic forces. Compared to static forces, where the computation and analysis of loads, moments, and other factors can be obtained conveniently, dynamic forces, on the other hand, pose a special concern for engineers as these forces produce oscillating or time-varying loads by moving structures around their equilibrium positions. To examine how mechanical systems react to these forces, the vibration testing technique is used.

During the 18th century, scientists like Euler and Bernoulli started exploring the oscillatory motion of beams and strings and began creating models to explain this behavior. This can be considered the start of the study of vibrations. Over time, these theoretical concepts and fundamental studies gave way to real-world applications such as the automotive industry, electronics, manufacturing, consumer products, defense, and especially in the aerospace industry, where vibration effects have a direct impact on the functionality and safety of aircraft structures. This process allows engineers to identify and address potential weaknesses before a product reaches the market or a machine fails in the field, which ultimately saves costs and protects human well-being.

In aerospace engineering, vibration testing is commonly used for analyzing the structural behavior of aircraft wings, fuselage panels, control surfaces, and engine mounts, which are all subject to dynamic forces during operation. Engineers can determine a structure's fundamental frequencies and mode shapes by conducting controlled vibration experiments. These factors show how a structure will react to actual operating situations, like uninstalled vs an installed engine drag or aerodynamic loading.

In this experiment, both short and long aluminum beams were excited at various frequencies using an electrodynamic shaker. A digital stroboscope, which makes a vibrating object appear stationary when the strobe flash frequency matches the vibration frequency, was then used to visualize the vibration behavior of the beams. This made it possible to observe nodes directly, which helped in determining the resonance frequency of each beam. After testing both structures, the beams were switched with aerospace structures, such as a rib and a model airplane structure.

There are many benefits of vibration testing that are very useful to engineers to determine the structural integrity of a structure. The ability to detect structural flaws, reducing maintenance costs through early detection of potential failures, and improving product reliability and safety are some of the advantages of this method of testing. However, it comes with its drawbacks as well, such as high costs for equipment and skilled personnel, complexity of test setup, possible interference from outside vibrations, and equipment calibration.

Despite the challenges, vibration testing is still an essential method in the evaluation of aeronautical structures and plays a fundamental role for new engineers to get exposed to experimental assessment of physical systems.

Equipment/Apparatus

Below is the list of equipment used during the experiment:

- Al-6061-T651 beam samples (long or short) and aircraft members
- Screws, washers, tube spacers, and Phillips head screwdriver
- Calipers, ruler, and tape measure
- Whiteboard markers
- ICP® (integrated-circuit piezoelectric) accelerometer and sensor signal conditioner
- Hot glue and gun
- Digital oscilloscope (V and s)
- Power supply amplifier
- Electrodynamic shaker and cooling fan
- Sweep sine generator (cycle/s)
- Digital stroboscope (flash/min) and tripod
- Various wires, cables, and connectors
- Power supply strip and possibly extension cord

Figure 1: Al-6061-T651 Beam Samples (Long and Short)

Figure 2: Aerospace Structures: Rib and Model Airplane

Figure 3: Screws, Washers, Tube Spacers, and Phillips Head Screwdriver

Figure 4: Calipers, Ruler, and Tape Measure

Figure 5: Digital Oscilloscope and Sweep Sine Generator

Figure 6: Digital Stroboscope with tripod, ICP® (integrated-circuit piezoelectric) accelerometer and sensor signal conditioner, and Digital Supply Amplifier

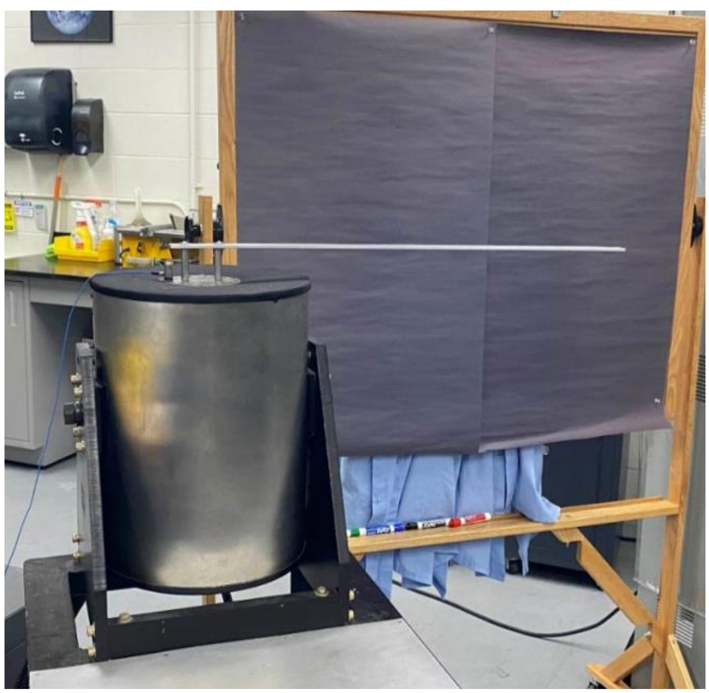

Figure 7: Electrodynamic Shaker

Method

1. Verify and power Test equipment
2. Measure short beam cross section
3. Connect short beam to shaker
4. Measure beam length from far end to first connection point
5. Set up power supply amplifier and sweep sine generator
6. Turn frequency of sweep sine generator up until beam resonates in mode 1
7. Observe and record resonance frequency on oscilloscope, generator and by using a digital stroboscope.
8. Tune frequency up and repeat step 7 for modes 2 and 3.
9. Turn off sine generator and replace short beam with long beam.
10. Repeat steps 5-8 for long beam along with finding mode 4.
11. Lower frequencies and turn off equipment.

Results

Table 1: Beam sizes (mm)

	L	h	b
Short	761	5.05	25.55
Long	1275	5.01	25.52

Table 2: Frequencies (Hz) for short beam:

	F_{SSG}	F_{piezo}	F_{strobe}
Mode 1	6.5	6.41	6.5
Mode 2	40.1	40.32	40.06
Mode 3	111.5	111.1	111.4

Table 3: Nodal distances (cm) for short beam:

	Node 1	Node 2	Node 3
Mode 1	N/a	N/a	N/a
Mode 2	59.3	N/a	N/a
Mode 3	37.1	65.8	N/a

Table 4: Frequencies (Hz) for long beam:

	F_{SSG}	F_{piezo}	F_{strobe}
Mode 1	2.4	2.4	2.4
Mode 2	14.5	14.49	14.58
Mode 3	40.9	40.98	40.9
Mode 4	80.1	80	80.066

Table 5: Nodal distances (cm) for long beam:

	Node 1	Node 2	Node 3
Mode 1	N/a	N/a	N/a
Mode 2	99.8	N/a	N/a
Mode 3	62.8	110.5	N/a
Mode 4	44.2	81.6	114.7

Analysis

The Theoretical Frequency was calculated using Equations 1-4. In the equations below: b is the base, h is the height, A is the cross-sectional area, I is the moment of inertia, L is the length, n is the mode number, λ_n is the eigenvalue of mode n, E is Young's Modulus, ρ is the density, ω_n is the angular frequency of mode n, and f_n is the theoretical frequency of the mode n.

$$A = bh \tag{1}$$

$$I = \frac{bh^3}{12} \tag{2}$$

Table 6: Calculated Parameters of Al-6061 Short and Long Beams

6061 Al Sample	Area (m^2)	Inertia (m^4)	Density (kg/m^3)	Young's Modulus (GPa)
Short Beam	$1.29*10^4$	$2.74*10^{-10}$	2700.00	68.90
Long Beam	$1.28*10^4$	$2.67*10^{-10}$	2700.00	68.90

Table 7: Eigenvalues of each Mode

	Mode 1	Mode 2	Mode 3	Mode 4
λL	1.88	4.69	7.85	11.00

Values from the Tables 6 and 7 were used in Equations 3 and 4 to calculate the Theoretical Frequencies of short and long beams and different modes.

$$\omega_n = \frac{(\lambda_n L)^2}{L^2}\left(\frac{EI}{\rho A}\right)^{\frac{1}{2}} \tag{3}$$

$$f_n = \frac{\omega_n}{2\pi} \tag{4}$$

Table 8: Theoretical Frequency of the Short and Long Beams

Theoretical Frequency (Hz)				
6061 Al Sample	Mode 1	Mode 2	Mode 3	Mode 4
Short Beam	7.15	44.5	124.72	N/A
Long Beam	2.53	15.7	44.1	86.55

By comparing the Experimental Frequencies from Tables 2 and 4 to the Theoretical Frequencies from Table 8, percentage difference was calculated using Equation 5 below. In the equation *a* is the experimental value and *b* is the theoretical value.

$$\%\,diff = \frac{|a-b|}{\left|\left(\frac{a-b}{2}\right)\right|} * 100\% \tag{5}$$

Table 9: Percentage difference of F_{SSG}, F_{piezo} and F_{strobe} vs Theoretical Value

Short	Mode 1	Mode 2	Mode 3	Mode 4
Long				
% diff of F_{SSG}	9.52	10.40	11.19	N/A
	5.27	7.95	7.53	7.74
% diff of F_{piezo}	10.91	9.86	11.55	N/A
	5.27	8.02	7.33	7.87
% diff F_{strobe}	9.52	10.50	11.28	N/A
	5.27	7.40	7.53	7.78

From Table 9, the percentage difference for the long beam was lower than the short beam for all the modes. For Mode 1, the short beam was ~9-11% off the theoretical value and the long beam was 5.27% off the theoretical value. For Mode 2, the short beam was ~9-10% off the theoretical value and the long beam was ~8% off the theoretical value. For Mode 3, the long beam was ~7.8% off the theoretical value.

By using the values from Table 9, the percentage difference of F_{SSG}, F_{piezo}, and F_{strobe} between each other was calculated using Equation 5. This time, *a* and *b* were replaced by different experimental frequencies.

Table 10: Percentage difference of F_{SSG}, F_{piezo} and F_{strobe} between each other

Short	Mode 1	Mode 2	Mode 3	Mode 4
Long				
% diff of F_{SSG}	13.61	5.39	3.15	N/A
and F_{piezo}	0.00	0.86	2.63	1.60
% diff of F_{piezo}	13.61	6.34	2.35	N/A
and F_{strobe}	0.00	8.02	2.63	1.05
% diff F_{strobe} and	0.00	0.95	0.80	N/A
F_{SSG}	0.00	7.16	0.00	0.55

As shown in Table 10, the highest percentage difference was between F_{SSG} and F_{piezo} in Mode 1 at 13.61% but by Mode 3 the percentage difference dropped to 3.15. It can be concluded that at low frequencies, the system is harder to measure accurately. The Piezo oscilloscope struggles at frequencies below 10 Hz. At higher frequencies, all the frequency measurement devices converge better and produce smaller percentage differences.

To calculate the percentage difference of measured nodal distance compared to theoretical values. The theoretical values were first calculated using the standard theoretical nodal distance ratios *x/L* found in the mode shapes of a cantilever beam. The ratios were multiplied by the measured length of the short and long beams.

Table 11: Theoretical Nodal Distance

	Mode 2	Mode 3		Mode 4		
x/L	0.783	0.504	0.868	0.358	0.644	0.906
Short Beam (mm)	595.86	383.54	660.55	N/A	N/A	N/A
Long Beam (mm)	998.33	642.60	1106.70	456.45	821.10	1155.15

Values from Tables 3, 5 and 11 were used in Equation 5 to calculate the percentage difference between measured and theoretical nodal distance values. In the equation *a* is the experimental value and *b* is the theoretical value.

Table 12: Percentage difference of Measured Nodal Distance vs Theoretical Nodal Distance

	Mode 2	Mode 3		Mode 4		
% diff	Node 1	Node 1	Node 2	Node 1	Node 2	Node 3
Short Beam (mm)	0.48	3.32	0.39	N/A	N/A	N/A
Long Beam (mm)	0.03	2.30	0.15	3.22	0.62	0.71

From Table 12, the node distances were measured accurately as the highest percentage difference is only 3.32% which can be accounted for the human error.

Conclusion

The results and analysis show that vibration testing can be a reliable way to determine an object's resonance frequency, unless the resonance frequency for that object is extremely low, as demonstrated by the dramatic jump in percent difference between the experimental values for mode 1, being over 13%, dropping down to less than 3% on average for modes 3 and 4. There is still the possibility that errors occurred during the experiment, and the lower frequencies made those errors more noticeable. A simple solution would be to repeat the experiment for those values using the same object, or a similar one. Looking at the experimental and theoretical frequencies, a trend emerges for the long beam regardless of which experimental value is used; the percent difference always remains around 5% for mode 1, increasing to over 7% for modes 2, 3, and 4. This trend does not appear for the long beam with a 9.5% difference for the strobe and sine wave generator, but almost 11% for the oscilloscope, pointing to a possible error during the experiment, which could warrant a second trial, this same trend continuing in all modes with the strobe light and sine wave generator being similar in values, while the oscilloscope is slightly off. One possible source of the error could be in the form of human error while operating the oscilloscope, since both the sine wave generator and the strobe lights have less human input when determining the experimental value vs, the oscilloscope, which required occasional adjustments during the experiment. As for the node values, considering that a 3.3% difference was the maximum, it can be safely concluded that node distances can be reliably determined using vibration testing.

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A. Bharadwaj: Introduction and Equipment
J. Nguyen Jr.: Abstract and Conclusion
R. Orr: Method and Results
K. Satvaldy: Analysis

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